

Output: Stria terminalis, which terminates mainly on mammillary & septal nuclei.

Functions: **emotional center of brain, fear, anger & sexual desire.**

Lecture (17): Cerebellum

Gross anatomy of Cerebellum:

- The cerebellum is situated in the **posterior cranial fossa** and is separated from the cerebrum by **tentorium cerebelli**.
- It is the largest part of the **hindbrain** and lies posterior to the fourth ventricle, the pons & the medulla oblongata.
- It consists of **two cerebellar hemispheres** joined by a median **vermis**, which is divided into:
 - 1- **Superior vermis**, present on the superior surface.
 - 2- **Inferior vermis** lies in the **vallecula** (a depression between cerebellar hemispheres on the inferior surface), and consists of **pyramid, uvula and nodule**.
- Three cerebellar peduncles connect each hemisphere to brainstem:
 - 1- The **Superior peduncle** connects with the midbrain.
 - 2- The **Middle peduncle** connects with the pons.
 - 3- The **Inferior peduncle** connects with the medulla.
- The surface of the cerebellum is indented by parallel **folia** separated by sulci.

■ Fissures of cerebellum:

- 1- **Primary fissure:** on upper surface, V-shaped, separates the anterior lobe from the middle lobe.
- 2- **Postero-lateral fissure** extends on inferior surface of cerebellum, separates the folliculonodular lobe from the main part of cerebelum.
- 3- **Horizontal fissure:** divides the cerebellum into superior and inferior surfaces, extends around the posterolateral border of each hemisphere.

Anatomical divisions (lobes) of cerebellum:

- 1- **Anterior lobe:** cranial lies to the primary fissure. on the superior surface of the cerebellum.
- 2- **Middle (posterior) lobe** is the largest part of the cerebellum, is situated between the primary fissure & the posterolateral fissure.
- 3- **Flocculonodular lobe:** lies caudal to the posterolateral fissure.

Functional & developmental divisions of cerebellum:

➤ **Vestibulocerebellum (archicerebellum):**

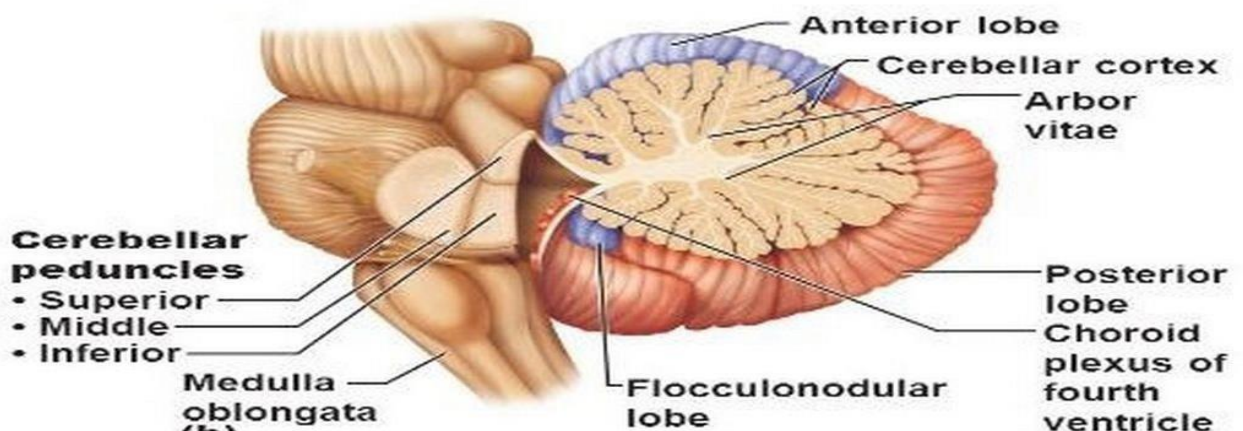
- It consists of Flocculo-nodular lobe.
- It receives from the vestibular system, superior colliculus & visual cortex.
- It projects to the vestibular nuclei.
- There is no deep cerebellar nuclei for archicerebellum.
- It is concerned with equilibrium, control of head & eye movement & smooth pursuit eye movements.

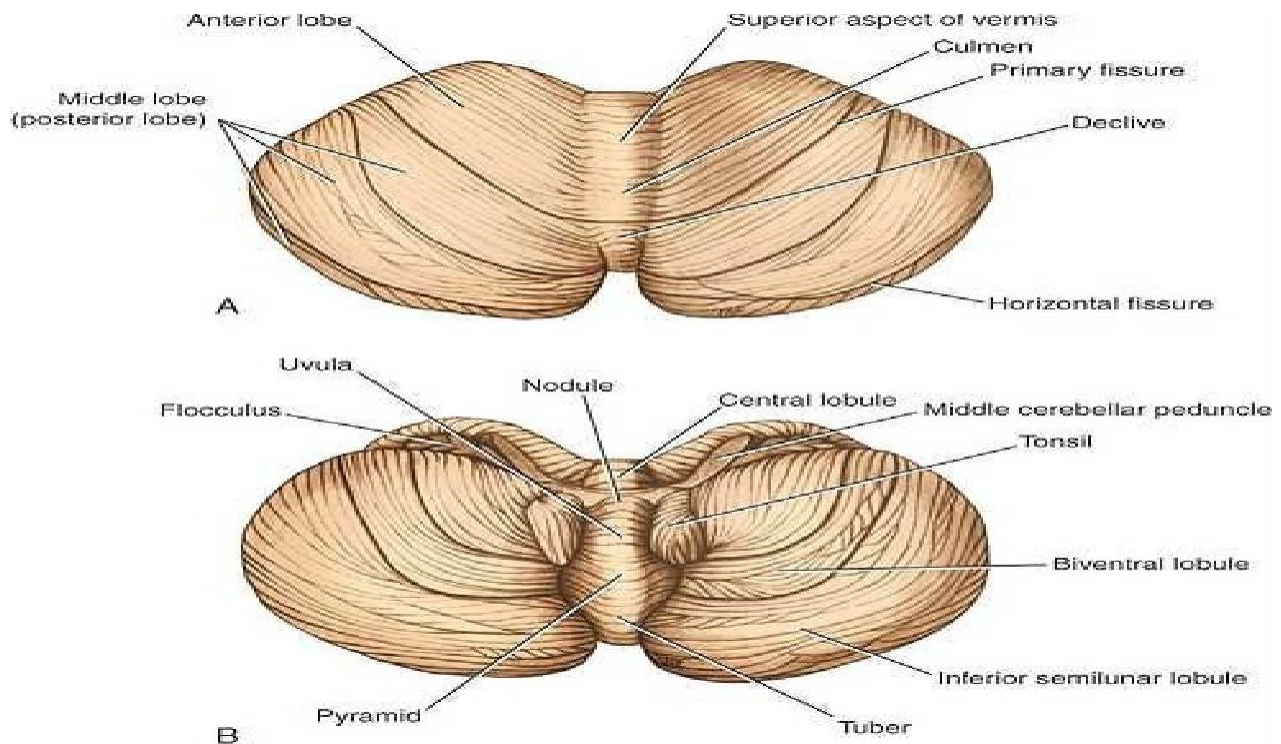
➤ **Spino-cerebellum (paleocerebellum):**

- It consists of the **vermal & paravermal zones of cerebellum.**
- It is projected to **Fastigial & interposed nuclei.**
- It relates to the **spinal cord.**
- It is concerned with **adjustment of muscle tone & coordination of ongoing movement.**

➤ **Cerebro-cerebellum (neocerebellum):**

- It consists of the greater lateral zone of cerebellar hemispheres.
- It projects to **dentate nucleus.**
- It relates to the **cerebral cortex.**
- It is concerned with **planning of intended movement, motor learning & cognitive function.**





(A)superior surface & (B) inferior surface.

Internal Structure of the Cerebellum

- ▣ **Cerebellar Cortex:** outer covering of grey matter. It is formed of **3** layers:

1- Molecular layer (external layer):

It is formed of **dendritic trees** of **Purkinje cells**, axons of granule cells & **stellate** and **basket** cells.

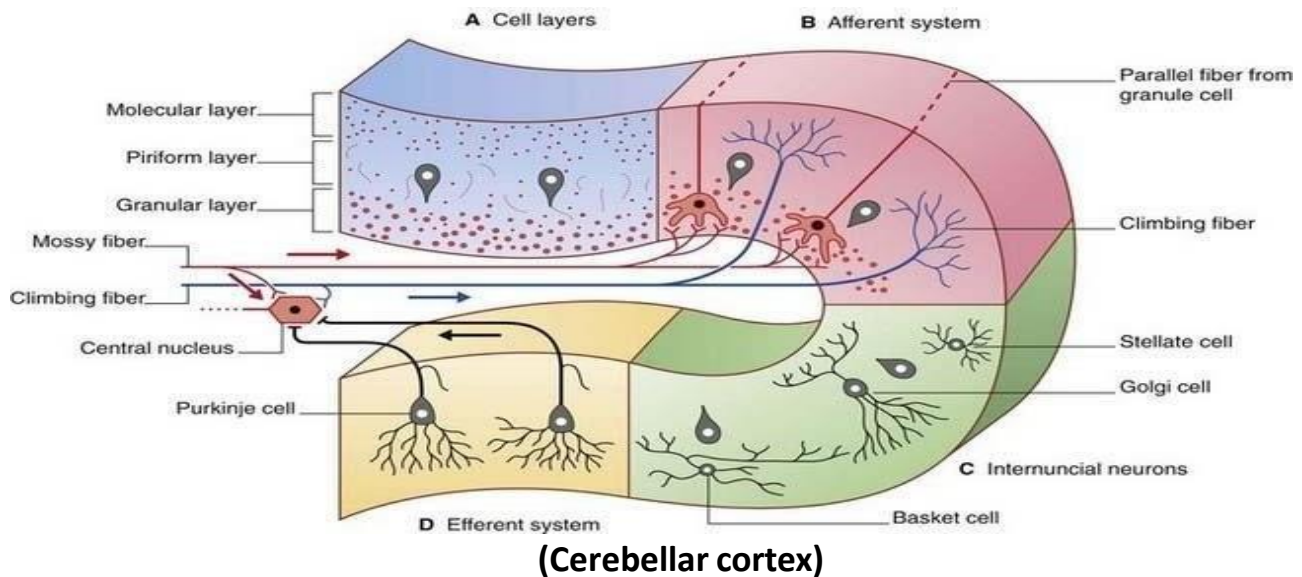
2- Purkinje cell layer (middle layer):

- It consists of a single layer of **large flask-shaped Purkinje cells**, their dendrites extend to molecular layer, while their axons project through the granular layer and synapse with deep cerebellar nuclei.
- It is the **only efferent pathway** from the cerebellar cortex.

3- Granule cell layer (internal layer): It contains granule and Golgi cells.

▣ **Inner white matter:**

- **Beneath** the cerebellar cortex, is formed of axons coursing to and from the cerebellar cortex.
- The **branching pattern** of white matter is called **arbor vitae cerebelli**.
- **Embedded** within the white matter of each hemisphere are **four intracerebellar nuclei**.



Intracerebellar nuclei:

- They are arranged from medial to lateral; **Fastigial, Globose, Emboliform & Dentate nuclei.**

- **Axons of deep nuclei form the output of the cerebellum.**

■ Dentate nucleus:

- It is the **largest** and **most lateral** intracerebellar nucleus.
- It has a crumpled bag-shaped, with the opening facing medially.
- For planning & control of movement.
- **Its input:** receives **cortico-ponto-cerebellar fibers** to lateral zone of cerebellar cortex (**neocerebellum**).
- **Its output:** projects to: **Ventral lateral nucleus of thalamus** projects to (areas 6 & 4) & **Red nucleus.**

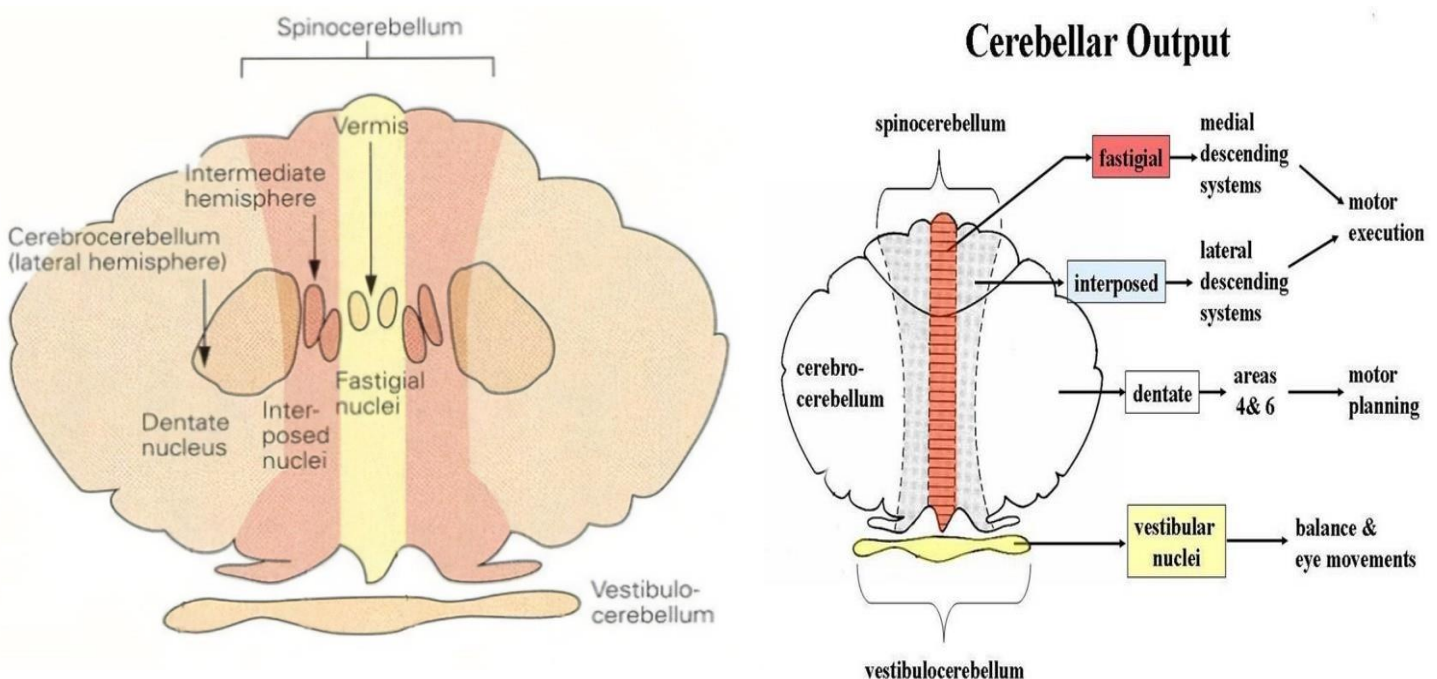
■ Interposed nuclei:

- They include **emboliform & globose nuclei.**
- For control of muscles of the limbs.
- **Its input:** receive from dorsal & ventral spinocerebellar tracts and cuneocerebellar tract, end on **paravermal zone of cerebellar cortex (paleocerebellum).**

- Its output: projects to: Ventral lateral nucleus of thalamus, Red nucleus & inferior olivary nucleus.

■ **Fastigial nucleus:**

- It is the **most medial intracerebellar nucleus**.
- For control of axial & girdle muscles (equilibrium).
- **Its input:** receive spinocerebellar and vestibular afferents, end on vermal zone of cerebellar cortex (**paleocerebellum**).
- **Its output:** projects to: **Vestibular nuclei** to spinal cord (vestibulospinal tracts). **Reticular formation** to spinal cord (reticulospinal tracts).



Connections of cerebellum:

- The input of cerebellum is mossy or climbing fibers:
 - 1- The **only climbing fibers** is the **olivo-cerebellar tract**.
 - 2- **All fibers entering the cerebellum** are **mossy** except olivo-cerebellar tract. Synapse firstly with granule cells (excitatory), then axons of granule cells (parallel fibers) synapse with purkinje cells.
- Output of cerebellar cortex via purkinje cells.
- Output of cerebellum through **deep cerebellar nuclei**.

▣ **Superior cerebellar peduncle (SCP):**

Afferents:

- **Anterior spinocerebellar tract** (from ventral spinal border cells).
- **Tecto-cerebellar fibers** (from tectum of midbrain).

Efferents:

- **Dentato-thalamic fibers** (Axons of dentate nucleus pass through superior cerebellar peduncle, end by synapsing with the contralateral ventrolateral nucleus of the thalamus, project to **area 6 & area 4**).
- **Dentato-rubral fibers** (to contralateral **red nucleus**).

▣ **Middle cerebellar peduncle (MCP):**

Afferents:

- **Cortico-ponto-cerebellar fibers:** (from **cerebral cortex** of all lobes to **pontine nuclei** in pons, which give rise to **transverse pontocerebellar fibers**, which cross the midline and enter the opposite the **middle cerebellar peduncle** to **granule cells** of cerebellar cortex (**Mossy fibers**), synapse with Purkinje cells of the cerebellar cortex.

▣ **Inferior cerebellar peduncle (ICP):**

Afferents:

- **Posterior spinocerebellar tract** (from Clarke's nucleus of posterior horn of spinal cord).
- **Cuneocerebellar tract** (from accessory cuneate nucleus in medulla).
- **Olivocerebellar tract** (which originate from inferior olivary nucleus of opposite side).
- **Vestibulocerebellar tract** (from vestibular nuclei).
- **Reticulocerebellar tract** (from reticular nuclei in pons & medulla).
- **Trigemino-cerebellar tract** (from sensory nuclei of trigeminal nerve).
- **Stria medullaris fibers** (from arcuate nucleus in medulla).

Efferents:

- **Cerebello-vestibular fibers** (from fastigial nucleus to vestibular nuclei).
- **Cerebello-reticular fibers** (from fastigial nucleus to reticular nuclei). -

Cerebell

o- olivary fibers.

Blood supply of cerebellum:

- 1- **Posterior inferior cerebellar artery:** arises from the vertebral artery, supplying the back of the inferior surface of cerebellum.
- 2- **Anterior inferior cerebellar artery:** arises from the basilar artery supplying **anterior part** of the **inferior surface** of cerebellum.

3- **Superior cerebellar artery:** arises from the **basilar artery**, supplying the superior surface of the cerebellum.

Lesion of cerebellum: ○ Ataxia is inaccurate movements. Hypotonia

- wide-based, lurching gait.
- Intention tremors.
- Decomposition of movement.
- Dysdiadochokinesia is the inability to perform alternating movements regularly and rapidly.
- Dysarthria: slurred, staccato speech (is the inability to articulate words properly).
- Nystagmus is a rhythmical oscillation of the eyes.
- Dysmetria: Overshooting or undershooting the target.

